

Spring

The Spring Multimodal Unified Framework:
A Training-Is-Audit Architecture for
Omni-Modal Large Models

SCX

202671

v1.0 — — SCX — Spring

Abstract

SPRING—— SPRING
(1) ——DFT —— (2) SPRING—— M_t (3) RAGSPRING (4) YAJIE M_t (5) CERCIS
SPRING “” SPRING→SPRING→ YAJIE→CERCIS—— GPTClaudeGemini $M = 1$ UNDECLARED
SPRINGborn audited——
SpringYajieCercis

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1

1.1

Large Language Models, LLMsGPT-2GPT-4ClaudeGemini “”

- (1)
- (2) GPT-4“”
- (3) “”——
—— ——

1.2 SCX

SCX
 $citescx_m l_a audit, scx_h amiltonian, yajie_p rotocolM=1$

- 1 SCX2 $M = 1$ ——“” “” “” $M = 1$
- 2 SCX1 $M > 1$ —— MM
- 3 -SCX3 ——

SPRING

1.3 Spring

SPRINGSCX——
 $citescx_spring_t rainerSpringYAJIEM_t$ “”
SPRING **Spring**——GPT “”

- DFT
-
-
-

Spring “ $\rightarrow \rightarrow \rightarrow \rightarrow$ ”

Definition 1.1 (Spring —). SPRING

$$Input \xrightarrow{\text{Enc}} Physics \xrightarrow{\text{SPRING}} Reasoning \xrightarrow{\text{YAJIE}} Audit \xrightarrow{\text{CERCIS}} Output$$

(1)

SPRING—————

1.4

- **Modality** — DFT $\mathbb{M} = \{m_1, m_2, \dots, m_K\}$
- **Physics Layer** SPRING — SPRINGDFT
- **Reasoning Layer** — RAG
- **Audit Layer** YAJIEM_tCERCIS
- **Large Model** —

1.5

- (1) 2SPRING —
- (2) 3 —
- (3) 4 — SPRING
- (4) 5 — LLMRAG
- (5) 6 — YAJIEM_t
- (6) 7 — DFT
- (7) 8 — CERCIS
- (8) 9 — GPT/Claude/Gemini $M = 1$ UNDECLARED
- (9) 10 —
- (10) 11

2 Spring

2.1

SPRING 1

DFT | MD | Text | Image | Audio | Video | Tabular

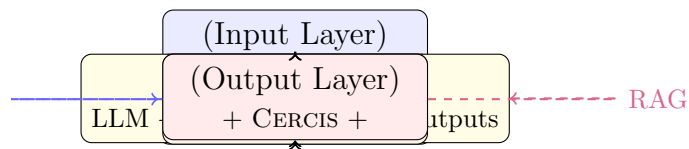


Figure 1: SPRING(1) (2) — RAG(3) +CERCIS+

2.2

1 **Training-Is-Audit** SPRING — $t V_{\text{consensus}}^{(t)} \mathbf{Y}_{\text{AJIE}} \mathbf{S}_{\text{AJIE}}^{(t)} \text{Lyapunov} \Psi_t M_t \text{CERCIS} S_{\text{CERCIS}}^{(t)}$ —

2 **Query-Is-Computation** — “” SPRING

3 **Output-Is-Audit** — $M_t \text{CERCIS}$

4 **Cross-Modality Verifiability** — DFT

5 **-Data-Parameter Binding** $M_t D_{\text{hash}}$

2.3

1SPRING

Table 1:				
	GPT-4	Claude 3	Gemini	Spring
	+	+	+++	DFT/MD/
				SPRING
=				
M	$M = 1$	$M = 1$	$M = 1$	$M_t > 1$
M	UNDECLARED	UNDECLARED	UNDECLARED	
				CERCIS
	RLHF	Constitutional AI		YAJIE

¹ $M = 1$ UNDECLARED—— SPRING

3

3.1

$\text{SPRINGEnc}_\theta : \mathbb{M} \rightarrow \mathbb{R}^d$

Definition 3.1. $\mathbb{M} = \{DFT, MD, Text, Image, Audio, Video, Tabular\}$

$$\text{Enc}_\theta(x, m) = W_{proj} \cdot \phi_m(x; \theta_m) \quad (2)$$

- $\phi_m(\cdot; \theta_m) : \mathcal{X}_m \rightarrow \mathbb{R}^{d_m} m \theta_m$
- $W_{proj} \in \mathbb{R}^{d \times d_m}$
- $\theta = \{\theta_m\}_{m \in \mathbb{M}} \cup \{W_{proj}\}$

3.2

—— ///CLIP WhisperViT

1 DFT $\text{DFT} \text{——} \{\mathbf{R}_i\}_{E_{\text{tot}}} \{\mathbf{F}_i\}_{\sigma} \text{DOS}(E) \text{——}$

- ACE
citeace
-
-

$$\phi_{\text{DFT}}(\{\mathbf{R}_i, E, \mathbf{F}, \sigma, \text{DOS}\}) = \text{ACE}(\{\mathbf{R}_i\}) \oplus \text{MLP}(E, \sigma) \oplus \text{KDE-Embed}(\text{DOS}) \quad (3)$$

$\oplus$

2 MD $[\mathbf{R}(t_1), \dots, \mathbf{R}(t_T)]$

$$\phi_{\text{MD}}(\mathbf{R}(t_1 : T)) = \text{Transformer}_{\text{temp}}(\text{ACE}(\mathbf{R}(t_1)), \dots, \text{ACE}(\mathbf{R}(t_T))) \quad (4)$$

$\text{Transformer}_{\text{temp}} \text{Transformer}$

3

$$\phi_{\text{Text}}(x) = \text{LLM-embed}(x) \quad (5)$$

BGEE5

4 /

$$\phi_{\text{Image}}(x) = \text{ViT}(x), \quad \phi_{\text{Video}}(x) = \text{TimeSFormer}(x) \quad (6)$$

$\text{Transformer-Transformer}$

5

$$\phi_{\text{Audio}}(x) = \text{Whisper-enc}(x) \quad (7)$$

Whisper

6

$$\phi_{\text{Tabular}}(x) = \text{FT-Transformer}(x) \quad (8)$$

-TransformerFeature Tokenizer Transformer

7

$$\phi_{\text{Graph}}(G) = \text{GNN}(G) \quad (9)$$

E(3)-GNN

3.3

Definition 3.2. $(x_m, x_{m'})mm'DFT$

$$\mathcal{L}_{align} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{Enc}_{\theta}(x_i^m, m)^{\top} \text{Enc}_{\theta}(x_i^{m'}, m')/\tau)}{\sum_{j=1}^B \exp(\text{Enc}_{\theta}(x_i^m, m)^{\top} \text{Enc}_{\theta}(x_j^{m'}, m')/\tau)} \quad (10)$$

$B\tau$

DFTMD “” SPRING — SPRING

3.4

DFT

$$\text{Route}(x, m) = \begin{cases} \text{Physics} \rightarrow \text{Reasoning} \rightarrow \text{Audit} & m \in \{\text{DFT}, \text{MD}, \text{Tabular_scientific}\} \\ \text{Reasoning} \rightarrow \text{Audit} & m \in \{\text{Text}, \text{Image}, \text{Audio}, \text{Video}\} \end{cases} \quad (11)$$

— 7

4 Spring

4.1

GPT-4ClaudeGemini—— “” “”

SPRING SPRING SPRING ——

4.2 Spring

SPRING

citescx_spring_trainer

Definition 4.1 (SPRING —). $\mathcal{D} = \{(\mathbf{R}_i, E_i, \mathbf{F}_i)\}_{i=1}^N$ DFT SPRING t

1 $\mathcal{D}M_t\{\mathcal{D}_m\}_{m=1}^{M_t}$

2 $\mathcal{D}_m f_m(\cdot; \theta_m)$ ——

3 **Yajie** $\{s_{\text{YAJIE}}(\mathbf{R}_i)\}_{i=1}^N$

4 **Lyapunov** $\Psi_t = \frac{1}{N} \sum_i \text{Var}_m[f_m(\mathbf{R}_i)]$

5 $s_{\text{YAJIE}}(\mathbf{R}_i) < \tau_{\text{noise}}$

6 $M_t \Sigma_0(\mathcal{D})\Psi_t M_{t+1}$

7 $V_{\text{consensus}}^{(t)} \mathbf{s}_{\text{YAJIE}}^{(t)} \Psi_t M_t^{\text{auto}} S_{\text{CERCIS}}^{(t)}$

SPRING[4]

Theorem 4.2 (— P1). $M_t \Delta$

$$\mathbb{P} \left(\bigcap_{m=1}^{M_t} |f_m(\mathbf{R}) - E_{\text{ref}}(\mathbf{R})| > \Delta \right) \leq \exp \left(-2M_t^{\text{eff}} \cdot \frac{\Delta^2}{\bar{\sigma}^2} \right) \quad (12)$$

$$M_t^{\text{eff}} = M_t / (1 + \bar{\rho}_t)$$

Theorem 4.3 (Yajie — P2). $\sigma_\epsilon > 0$

$$\mathbb{E}[s_{\text{YAJIE}} \mid \sigma_\epsilon > 0] = \left(1 + \frac{\sigma_\epsilon^2}{\bar{\sigma}_{\text{expert}}^2} \right)^{-M_t/2} \quad (13)$$

M_t

Theorem 4.4 (Lyapunov — P3). $\{\Psi_t\}_{t \geq 0}$ SPRING Lyapunov

$$\Psi_{t+1} \leq \Psi_t, \quad \lim_{t \rightarrow \infty} \Psi_t = 0 \quad (14)$$

4.3

Definition 4.5. *RAG*

- (i) $\text{DFT_SINGLEPOINT}(\mathbf{R}, \text{settings}) \mapsto (E, \mathbf{F}, \sigma, \text{DOS}) \ \mathbf{R} \text{DFTSPRING}$
- (ii) $\text{POTENTIAL_QUERY}(\mathbf{R}, V_{\text{consensus}}) \mapsto (E_{\text{pred}}, \mathbf{F}_{\text{pred}}, s_{\text{YAJIE}}) \ \text{SPRING} V_{\text{consensus}} \mathbf{R}$
- (iii) $\text{MD_RUN}(V_{\text{consensus}}, \mathbf{R}_0, T, \Delta t, N_{\text{steps}}) \mapsto [\mathbf{R}(t)] \ \text{SPRING} T N_{\text{steps}}$
- (iv) $\text{PROPERTY_COMPUTE}(V_{\text{consensus}}, \text{property_type}) \mapsto \text{value} \text{ ---}$
- (v) $\text{POTENTIAL_TRAIN}(\mathcal{D}_{\text{new}}) \mapsto V_{\text{consensus}}^{\text{new}} \ \mathcal{D}_{\text{new}} \text{SPRING}$
- YAJIE M_t CERCIS

4.4

SPRINGAudit Anchor

---Ground TruthDFT

SCX

$\text{citesc} x_h \text{amiltonianSPRINGH}(\theta)$

---“”

Remark 4.6. (1) **DFT** k---

(2) **Spring** CERCIS N_{cov}

$N_{\text{cov}} < \tau$

5 LLMRAG

5.1

SPRINGLLM SPRINGLLM

(1) **Query Compiler**

“” POTENTIAL_QUERYPROPERTY_COMPUTE

(2) **Result Interpreter**

Definition 5.1 (SPRING-). q

1 $LLMq(intent, entities, constraints)$ $intent \in \{PropertyQuery, SimulationRequest, ComparisonRequest,$

2 $\mathcal{G}_{phys}$

3 $\mathcal{G}_{phys}\mathcal{R}_{phys}\mathcal{A}_{phys}$

4 **RAG** SPRING $q\mathcal{R}_{phys}$ ——

5 $LLM\mathcal{R}_{phys}RAG$

6 6

7 8

5.2 RAGSpring

SPRINGRAG

(a) POTENTIAL_QUERYPROPERTY_COMPUTE

(b) **Spring** —— CERCIS M_t

(c) —— “” $C_{44} > X$

(d) —— SPRING

RAG—— CERCIS

5.3 LLM

LLMSPRINGLLM

1 LLM

POTENTIAL_QUERYPROPERTY_COMPUTE

2 YAJIECERCIS ——LLM

3 LLM——

4 LLM LLMSPRING “”

LLM“”“”——

5.4

“”

Example 5.2. “AlN”

1 — LLM

- intent = PropertyQuery
- entities = {material : AlN}
- properties_requested = {chemical_stability, reactivity, bond_type}

2 —

1. Potential_Query(AlN_equilibrium_structure, V_AlN_consensus)
→ E_coh, F_eq
2. Property_Compute(V_AlN_consensus, "elastic_constants")
→ C_ij
3. Property_Compute(V_AlN_consensus, "phonon_dispersion")
→ (q)
4. RAG_Search("AlN chemical properties", filter: audited)
→ historical_results

3 — LLM

“SPRINGV_AlNM_t = 8CERCIS S = 1.04 —— (1) E_{coh} = -11.2 eV/atomYAJIE
0.94 (2) C₄₄ = 1.97 GPaBorn (3) —— A”

—— “E_{coh}”POTENTIAL_QUERY “”PROPERTY_COMPUTE “”PROPERTY_COMPUTE LLM

6 Yajie M_t

6.1

SPRING——RAG “”

6.2 Yajie

YAJIE

citeyajie_pprotocolSPRING

Definition 6.1 (YAJIE). $q\mathcal{C} = \{c_1, \dots, c_K\}$ $c_k M_k$

- SPRING $M_t c_k$
- c_k
- $/c_k$
- c_k

c_k YAJIE

$$s_{\text{YAJIE}}(c_k) = \exp \left(-\frac{1}{M_k} \sum_{m=1}^{M_k} \left(\frac{\hat{a}_m(c_k) - \bar{a}(c_k)}{\sigma_{\text{calib},m}} \right)^2 \right) \quad (15)$$

$\hat{a}_m(c_k) \in [0, 1] m c_k$ “” $1=\theta=\bar{a}(c_k)$ $\sigma_{\text{calib},m} m$

6.3 M_t

SPRING M_t —— M_t $\Sigma_0(\mathcal{D})$ M_t

Definition 6.2 (M_t). $qM_t(q)$

$$M_t^{\text{auto}}(q) = \Xi(\text{hash}(q, \mathcal{C}); \Psi_{\text{prev}}, \mathbf{s}_{\text{YAJIE}}^{(\text{prev})}, \Sigma_0(\text{context})) \quad (16)$$

$\text{hash}(q, \mathcal{C})$ —— M_t -

“” $M_t = 3$ DFT $M_t = 12+$

6.4

V1 V1 — Format Compliance

- RAG
- CERCIS
- M_t

——V1

V2 V2 — Consensus Consistency YAJIE

- $s_{\text{YAJIE}}(c_k) \geq \tau_{\text{consensus}} c_k$
- ———

V3 V3 — Cross-Modal Consistency

-
-
-

7

V4 V4 — Historical Consistency SPRING

-
- iiiiii

6.5

—————

Protocol 6.3. t

1 $\mathcal{R}_{\text{draft}}^{(t)} \mathcal{C}^{(t)}$

2 **Audit-V1**

3 **Audit-V2** $\mathcal{C}^{(t)}_{\text{YAJIE}} \mathcal{C}_{\text{low}}^{(t)} = \{c_k : s_{\text{YAJIE}}(c_k) < \tau_{\text{consensus}}\}$

4 **Audit-V3**—— RAG

5 **Audit-V4**

6 V1-V4—— CERCIS

7

- $\rightarrow$
- $\rightarrow$
- SPRING

8 $\mathcal{R}_{\text{draft}}^{(t)}$

7

7.1

SPRING DFT

Definition 7.1. $\mathcal{C}_{text} \mathcal{V}_{phys} \Phi : \mathcal{C}_{text} \rightarrow \mathcal{V}_{phys}$

$$\Phi(c) = v \in \mathcal{V}_{phys} \mid v \models c \quad (17)$$

c Cross-Modally Auditable

“” ——— “AlN6.2 eV” ——— DFTSPRING

Theorem 7.2 (— CM1). $cvM_t \geq 3v \bar{\sigma}^2 \delta = |claimed(v) - true(v)| \mid \delta > \Delta$

$$\mathbb{P}(detect \mid \delta > \Delta) \geq 1 - \exp \left(-2M_t^{eff} \cdot \frac{(\delta - \Delta)^2}{\bar{\sigma}^2} \right) \quad (18)$$

Proof. $vM_t H_0 : v = v_{claimed}$ vs $H_1 : v \neq v_{claimed}$ $M_tv\hat{v}_m T = \frac{1}{M_t} \sum_m (\hat{v}_m - v_{claimed})^2 / \bar{\sigma}^2$
 $H_0 T \sim \chi_{M_t^{eff}}^2 \mid H_1 \delta > \Delta T \lambda = M_t^{eff} \cdot \delta^2 / \bar{\sigma}^2 \chi^2(18) \quad \square \quad \square$

Corollary 7.3. $M_t M_t^{eff} = 6\delta = 3\bar{\sigma} \geq 1 - 10^{-15}$ ———

7.2

1 Direct Verification *c*

Example 7.4. “AlNWurtzite_a = 3.11 Å” POTENTIAL_QUERY(AlN_wurtzite, $V_{consensus}$) $\rightarrow$
 $a_{pred} \mid a_{pred} - 3.11 \mid > 3\sigma_{calib} \rightarrow$

2 Consistency Verification *c*

Example 7.5. “AlN” MD_RUN($V_{consensus}, \mathbf{R}_{AlN}, T = 2000K, \dots$) MD2000K $\rightarrow$

3 Multi-Expert Disagreement

Example 7.6. “AlNSiC” PROPERTY_COMPUTE($V_{AlN}, elastic$) $\rightarrow C_{ij}^{AlN}$ PROPERTY_COMPUTE(V_{SiC}, el)
 C_{ij}^{SiC} “”

7.3

-
- -
 - ———

• “”

Proposition 7.7. $\text{SPRING}_{\hat{v}_{phys}LLM} c \mid q = 0 \text{ LLM}$

8 Cercis

8.1

SPRING

$$\boxed{\text{Output}(q) \mapsto \left(\underbrace{\mathcal{R}_{\text{answer}}}_{\text{Cercis}}, \underbrace{S_{\text{CERCIS}}}_{\text{Cercis}}, \underbrace{\mathcal{T}_{\text{audit}}}_{\text{Cercis}} \right)} \quad (19)$$

Definition 8.1. (1) $\mathcal{R}_{\text{answer}} \xrightarrow{\text{YAJIE}} \text{RAG}$

(2) **Cercis** S_{Cercis}

$$S_{\text{CERCIS}}(\mathcal{R}_{\text{answer}}) = Q_{\text{prec}}(\mathcal{R}_{\text{answer}}) + \eta \cdot N_{\text{cov}}(\mathcal{R}_{\text{answer}}) \quad (20)$$

- $Q_{\text{prec}} \in [0, 1] \xrightarrow{\text{YAJIE}}$
- $N_{\text{cov}} \in [0, 1] \xrightarrow{cM_{\min} = 3}$
- $\eta = 0.2 \xrightarrow{\quad}$

(3) $\mathcal{T}_{\text{audit}}$

- q
-
-
- $V1-V4 \xrightarrow{\quad}$
- M_t
-

8.2 Cercis

CERCIS

$\text{citescx}_m l_a \text{audit}$ SPRING

Definition 8.2 (CERCIS). $\mathcal{R}_{\text{answer}} K\{c_1, \dots, c_K\} \xrightarrow{K_{\text{auditable}} \geq 3} \text{CERCIS}$

$$S_{\text{CERCIS}} = \underbrace{\frac{1}{K} \sum_{k=1}^K s_{\text{YAJIE}}(c_k)}_{Q_{\text{prec}}} + \eta \cdot \underbrace{\frac{K_{\text{auditable}}}{K}}_{N_{\text{cov}}} \quad (21)$$

CERCIS

- $Q_{\text{prec}} \xrightarrow{\quad}$
- $N_{\text{cov}} \xrightarrow{\quad}$
- $S_{\text{CERCIS}} \xrightarrow{N_{\text{cov}}} Q_{\text{prec}}$

8.3

$\mathcal{T}_{\text{audit}}$

(1) $\mathcal{T}_{\text{audit}}$

(2) ———

(3) $M_t D_{\text{hash}}$

SPRING “”

9

9.1 $M = 1$ UNDECLARED

——GPT-4Claude 3Gemini—— $M = 1$ UNDECLARED

1 GPT-4OpenAI GPT-4 $M = 1$ ——

- GPT-4MoEMixture of Experts “”
- GPT-4 “” “” “”
- GPT-4WebText
-

2 Claude 3Anthropic Claude 3Constitutional AIAIM = 1

- Constitutional AI——
- “”
- GPT-4Claude 3

3 GeminiGoogle Gemini

- $M = 1$ ——
- “” “”
- GPT-4——UNDECLARED

Definition 9.1 (UNDECLARED). *UNDECLARED*

- (i) M
- (ii)
- (iii)
- (iv)

GPT-4Claude 3GeminiUNDECLARED

9.2 Born AuditedSpring

SPRINGBorn Audited

Definition 9.2 (Born Audited). *Born Audited*

- (i) —— M_t YAJIECERCIS
- (ii) —— CERCIS M_t
- (iii) *SCX1*
- (iv)
- (v) —— M_t

2SPRING

Table 2:				
	GPT-4	Claude 3	Gemini	Spring
	UNDECLARED	UNDECLARED	UNDECLARED	Born Audited
M	$M = 1$	$M = 1$	$M = 1$	$M_t > 1$
=				
	CERCIS			
	SPRING			
M				

9.3 Born Audited vs. UNDECLARED

UNDECLAREDBorn Audited“” “” ——

- (1) UNDECLARED——
Born Audited——
- (2) UNDECLARED—— SCX2M = 1 Born AuditedM_t > 1
- (3) UNDECLARED——
Born AuditedSPRING——
- (4) UNDECLARED——
Born Audited——

10

10.1 Spring

SPRING

(1)

- —
- —
- — DFT
- **AI** — AI

(2)

- —
- NPC — SPRING
- —

(3)

- SPRING “” — $M_t = 3V_1 + V_2$
- —

10.2

SPRING

(1) — **Spring** M_t [4] $M_t\mathcal{O}(M_t)$ FLOP —

(2) — **RAG** RAG SPRINGLLM

(3) — **V1-V4** V_1V_2 — $V_2 V_3 V_4$ RAG

SPRINGGPT-42×5×

10.3 Spring

Spring

H1 SPRINGSPRING DFT — SPRING

H2 7.1 Φ —

“AlNSiC”

H3 LLM LLM5.1 — LLM SPRING

H4 M_t P_3M_t —

H5 SPRING——

H6 Cercis $\eta = 0.2\text{ML}$
citescx_ml_audit η —— -

H7 ——YAJIERAG——

10.4 SCX

SPRINGSCX SCX

- **SCX1** —— M_t
- **SCX2** $M = 1$
- **SCX3-** YAJIE
- **Yajie**
citeyajie_pprotocol
- **SCX**
citescx_hamiltonian**Spring**Lyapunov
- **Spring**
citescx_sspring_trainer
- **Cercis**
citescx_ml_audit
- **Galois-SCX**
citescx_galois||
- **SCX**
citescx_causal

11

SPRING——

(1) SPRING——

(2) $\rightarrow$ SPRING $\rightarrow$ LLM + RAG $\rightarrow$ YAJIE $M_t \rightarrow$ +CERCIS+—— “” “” “”

(3) SPRING ——

(4) CM1 M_t ——

(5) **Born Audited vs. UNDECLARED** —— Born AuditedSPRINGvs. UNDECLAREDGPT-4Claude 3Gemini UNDECLAREDBorn Audited

(6) SPRING+CERCIS+——

(7) SPRING—— LLM M_t η SPRING

SPRING ——

“”SPRING $\rightarrow \rightarrow$ YAJIE $\rightarrow$ CERCIS——

SPRING——**Born Audited, not UNDECLARED**——

SCXSCXYAJIECERCIS SPRING SCX

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A A

2SPRING

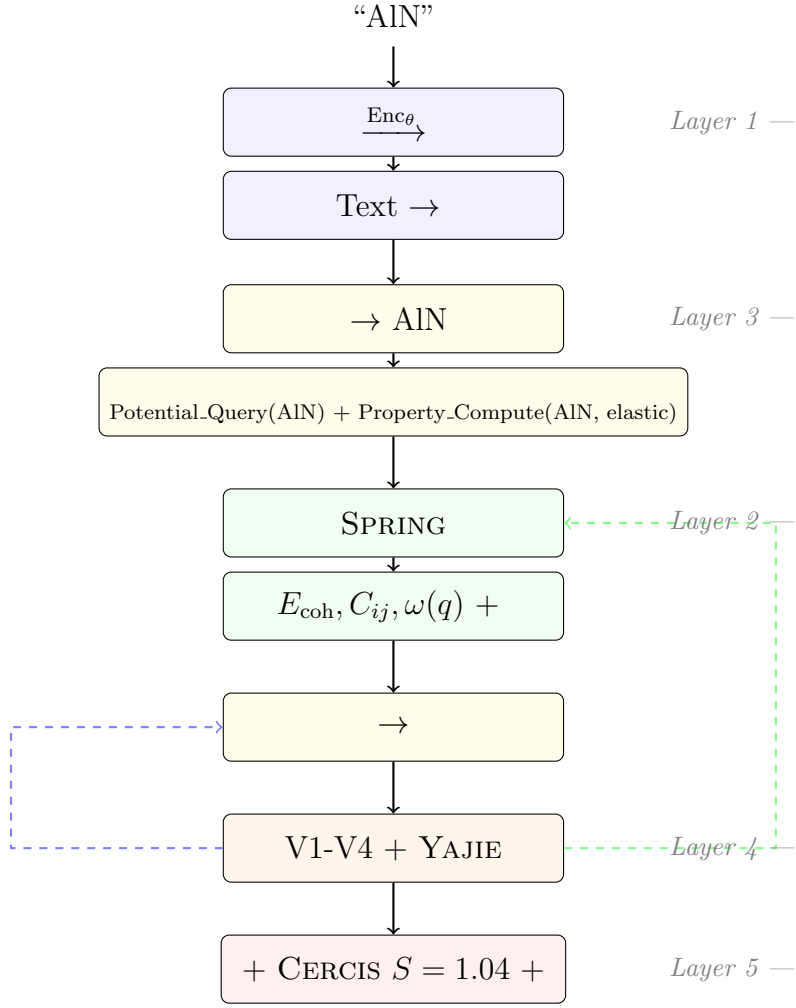


Figure 2: SPRINGAIN

B B

Table 3:

			/
$\mathbb{M}$			$\{\text{DFT, MD, Text, Image, Audio, Video, Tabular}\}$
Enc_θ			$\rightarrow \mathbb{R}^d$
ϕ_m			m
$V_{\text{consensus}}$	SPRING		M_t
M_t / M_t^{auto}			
s_{YAJIE}	YAJIE		$[0, 1]$
Ψ_t	Lyapunov		
S_{CERCIS}	CERCIS		$Q_{\text{prec}} + \eta \cdot N_{\text{cov}}$
Q_{prec}			YAJIE
N_{cov}			
$\mathcal{T}_{\text{audit}}$			
$\mathcal{H}(\mathcal{D})$			SHA-256
$\Sigma_0(\mathcal{D})$			